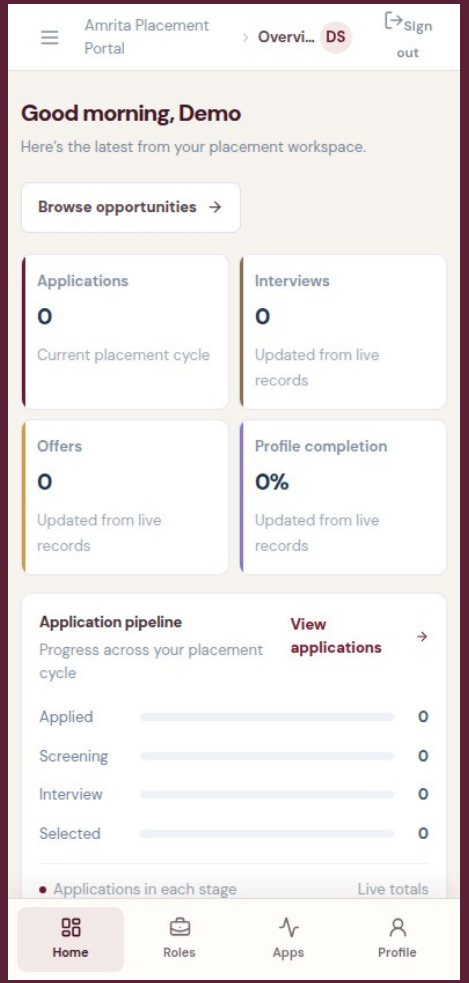
AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

BENGALURU CAMPUS

PLACEMENT WORK, IN ONE PORTAL.

Amrita Placement Portal · Bengaluru Campus

Student, recruiter and placement-team workflows connected to one Oracle-backed system.

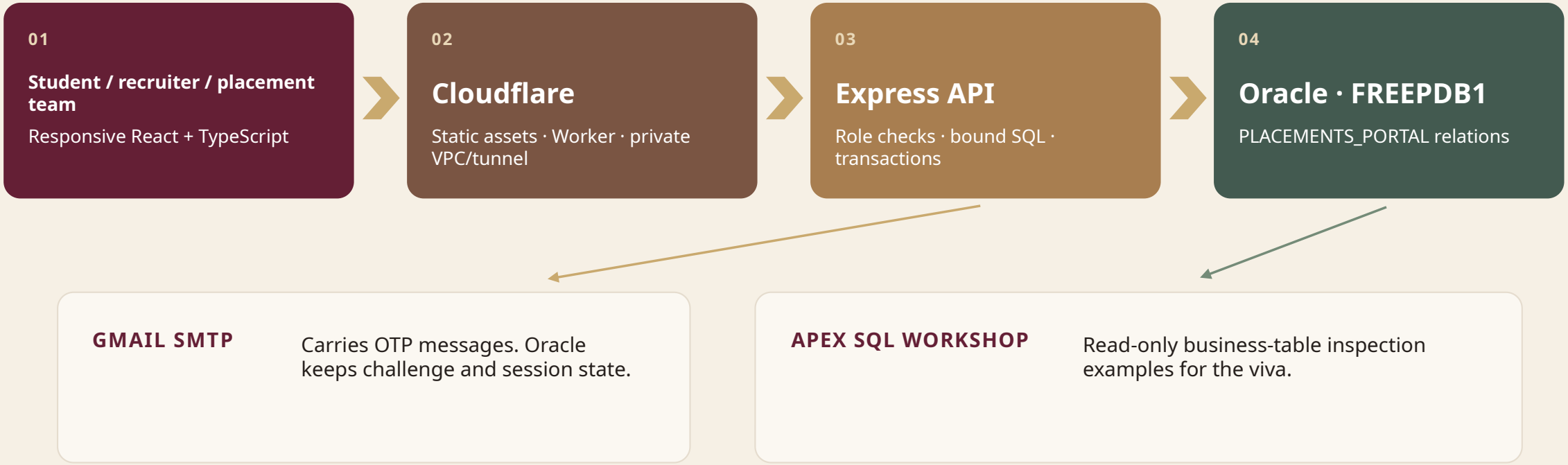


Synthetic Demo Student · QA capture

DBMS VIVA / PRODUCT + IMPLEMENTATION

One request path. Clear boundaries.

The portal UI, API and Oracle remain distinct; email and SQL Workshop sit on separate paths.



Source: README.md · docs/team-viva/DBMS-WALKTHROUGH.md · server/src/ · cloudflare/

The ER model becomes working relations.

Keys and constraints protect the workflow while API joins shape the views.

18 TABLES

[DERIVED] · counted from CREATE TABLE in schema DDL

IDENTITY

DEPARTMENTS
COMPANIES
USERS
STUDENTS
STUDENT_PHONE_NUMBERS

PLACEMENT

OPPORTUNITIES
OPPORTUNITY_ELIGIBLE_DEPARTMENTS
OPPORTUNITY_SKILLS
APPLICATIONS
PLACEMENTS
INTERNSHIPS

ENGAGEMENT

HACKATHONS
HACKATHON_REGISTRATIONS
CERTIFICATES

PLATFORM

OTP_EMAIL_LOCKS
OTP_CHALLENGES
SESSIONS
UPLOADS

Phones and opportunity eligibility/skills have relation rows. JSON arrays remain serialization mirrors; placement requirements also retain a historical snapshot.

INTEGRITY + SQL + TRANSACTIONS

PK / FK / UNIQUE / CHECK protect links, participation and valid values.
JOIN: Applications → Students → Opportunities → Companies · GROUP BY status · COUNT(*)
Profile + student + phone replacement commits or rolls back together in the API.

Count method: `rg -c "CREATE TABLE" database/001_schema.sql` → 18 [DERIVED]. Source: database/001_schema.sql · database/SCHEMA.md · server/src/db.ts

From opportunity to outcome.

The same Oracle-backed workflow supports students, recruiters and the placement team.

01 **Student**

Profile → eligible opportunity → application

02 **Recruiter**

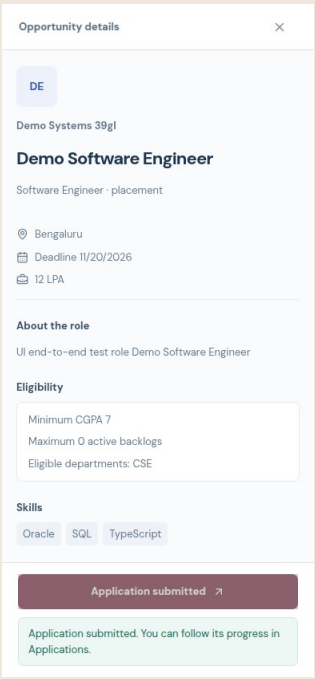
Company opportunity → application stage

03 **Placement team**

Approval → outcomes and reports

04 **Student team**

Hackathon registration → team list



Synthetic Demo Student · apply flow

QA counts [MEASURED]: 29 brand · 35 shell · 43 flows · 20 device + privacy

Layout [MEASURED]: 12 CSS widths; 200% text-size emulation at four widths. Browser checks cover exercised journeys, not every validation case.

QA [MEASURED]: public-device-results.json · public-qa-followup.md

AMRITA VISHWA VIDYAPEETHAM
BENGALURU CAMPUS

VALIDATE THE MODEL. TRACE THE WORKFLOW.

Follow one record through relational rules, SQL and an atomic API operation.

01

INTEGRITY

Primary and foreign keys connect the student, opportunity and application records. UNIQUE and CHECK rules reject invalid combinations.

02

QUERY

Join applications to students, opportunities and companies. Group by status to explain a report result.

03

TRANSACTION

A profile update changes related user, student and phone rows together—or rolls the operation back.

DEMO PATH student application → recruiter stage update → APEX read-only query → team handoff

Use synthetic demo records. APEX is for read-only inspection; application writes go through the API.